

A General Framework for Efficient Medical Image Analysis via Shared Attention Vision Transformer

Yihang Liu¹, Ying Wen², *Member, IEEE*, Longzhen Yang³, Lianghua He⁴, *Member, IEEE*, and MengChu Zhou⁵, *Fellow, IEEE*

Abstract—Vision Transformers (ViTs) demonstrate significant promise in medical image analysis but face two critical challenges: 1) their limited ability to capture local features in data-scarce scenarios, leading to data inefficiency, and 2) their high computational and storage demands of the full fine-tuning process in transfer learning, resulting in parameter inefficiency. To achieve efficient and accurate medical image analysis, we propose Shared Attention Vision Transformer (SAViT) that comprises three innovative modules: i) Shared Prior Attention (SPA) that enhances data efficiency by innovatively employing a visual prompt to sequentially share consistent attention weights across local image regions, thereby enabling the learning of translational invariance to capture locality; ii) MixPool that preserves global modeling ability by aggregating local features after SPA through a multi-pooling mechanism, thus effectively facilitating long-range dependency across local image regions; and iii) Low-rank Multi-head Self-Attention (Lr-MSA) that improves parameter efficiency by using low-rank weights of multi-head self-attention, hence reducing computational complexity while maintaining accuracy in medical image analysis. SAViT demonstrates strong generalization across multiple medical imaging modalities,

including retinopathy, dermoscopy, and radiography. Extensive experiments are conducted. The results indicate its high data efficiency and outstanding performance in comparison with more than 20 medical-specific and ViT-based models when all of them are trained from scratch. It excels in parameter-efficient tuning by surpassing 17 models across 6 datasets in transfer learning, with only 0.17M/0.23M trainable parameters on ViT-B/SwinViT-B backbones requiring 86.60M/88.00M parameters. Source code can be found at: <https://github.com/LYH-hh/SAViT>

Index Terms—Medical image analysis, prompt learning, data efficiency, parameter efficiency.

I. INTRODUCTION

VISION Transformers (ViTs) [1] have demonstrated strong capabilities in medical image analysis, particularly in capturing global anatomical relationships [2]. However, they suffer from a critical limitation: **data inefficiency**. Unlike Convolutional Neural Networks (CNNs), which inherently encode locality and translation invariance, ViT-based models typically require large-scale training datasets (often millions of images) to learn these inductive biases [3], [4]. These biases are essential for capturing fine-grained details necessary for accurate diagnosis of subtle diseases [2], [5], [6]. Unfortunately, in the medical domain, labeled datasets are scarce due to high cost, much time requirements, and much expertise required for manual annotation. As a result, ViTs usually underperform when trained from scratch on medical data, limiting their effectiveness in real-world clinical applications according to [7] and [8].

Traditional methods improve the data efficiency of ViTs in medical imaging by incorporating CNNs as complementary modules [7], [9] or employing ImageNet-pretrained ViT weights through transfer learning [2], [10]. While these methods offer certain advantages, they exhibit significant limitations: CNN-ViT hybrids typically suffer from limited generalizability across diverse medical imaging modalities (e.g., retinopathy, dermoscopy, and radiography), and transfer learning is hindered by **parameter inefficiency** due to domain shifts between medical images and others [2].

Recently, visual prompt learning has emerged as a promising technique for parameter-efficient transfer learning [11], [12]. By introducing a few learnable parameters within the

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Yihang Liu, Longzhen Yang, and Lianghua He are with the School of Computer Science and Technique, Tongji University, Shanghai 201804, China (e-mail: 2111131@tongji.edu.cn; yanglongzhen@tongji.edu.cn; helianghua@tongji.edu.cn).

Ying Wen is with the School of Communications and Electronic Engineering, East China Normal University, Shanghai 200241, China (e-mail: ywen@cs.ecnu.edu.cn).

MengChu Zhou is with the Tongxiang Institute of Artificial General, Tongxiang, Zhejiang 314511, China, and also with the Helen and John C. Hartmann Department of Electrical and Computer Engineering, New Jersey Institute of Technology, Newark, NJ 07102 USA (e-mail: zhou@njit.edu).

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latent feature space of a frozen pre-trained model backbone, visual prompts have improved task-specific performance in natural images [13], [14] and some disease-specific medical applications [15], [16], [17]. Building on the success of visual prompts in refining feature information without architectural modifications, this work investigates *how to enhance both data and parameter efficiency in ViTs by adapting a visual prompt—applicable to diverse medical modalities—to extract local medical features, particularly translation invariance, from global anatomical relationships captured by ViTs.*

To answer it, we propose a general framework, named **Shared Attention Vision Transformer (SAViT)**. It explicitly adapts a visual prompt to improve the efficiency of ViTs in medical image analysis. It has three modules: Shared Prior Attention (SPA), MixPool, and Low-rank Multi-head Self-Attention (Lr-MSA). The first two modules aim to enhance local feature extraction while preserving the global representation capabilities of ViTs, thus **enabling data efficiency on limited medical images**. The last one optimizes the self-attention computation specifically for medical image analysis, thereby resulting in **improved parameter efficiency**.

SPA improves locality by sharing region-invariant attention weights across local anatomical regions using the same ViT model. This design is inspired by the spatial consistency of anatomical structures in medical imaging [18], [19], where information from one anatomical region can provide valuable prior knowledge for others due to similar structural textures across anatomies [20]. Based on this insight, SPA innovatively employs a learnable visual embedding, termed **Anatomical-assisted Prompt (Anat-Prompt)**, to sequentially process local image regions under a circular feature interaction strategy. Specifically, this prompt iteratively accumulates anatomical information from adjacent regions and propagates this information as prior knowledge across anatomical structures via cross-attention, thereby enhancing the translational invariance of locality. **MixPool**, on the other hand, is introduced to preserve a global representation of the entire image in ViTs. Inspired by [21], it employs multi-scale AvgPooling to extract hierarchical global features from the output of SPA, establishing long-range dependencies across local anatomical regions. Therefore, by effectively integrating both local and global information through SPA and MixPool, SAViT achieves data efficiency in medical image analysis. To improve parameter efficiency, **Lr-MSA** is proposed to refine the traditional MSA [1] by novelly applying low-rank decomposition to the attention matrix. Specifically, it reduces the dimensionality of query and key matrices during attention computation while maintaining the original dimensionality of the value matrix, thus ensuring the rich representations of fine-grained anatomical structures with much fewer parameters than those needed by [1] and [22].

We evaluate SAViT under two training schemes: *Training from Scratch* and *Transfer Learning*, promoting its data efficiency and parameter efficiency, respectively. For the former, we adopt the plain ViT-S/B [1] as the backbone, forming the data-efficient variants: SAViT-S/B. For the latter, we employ SAViT on frozen ViT-B and SwinViT-B [22] by using Lr-MSA as trainable adapters to mitigate domain shifts, thereby yielding two parameter-efficient models: SAViT and

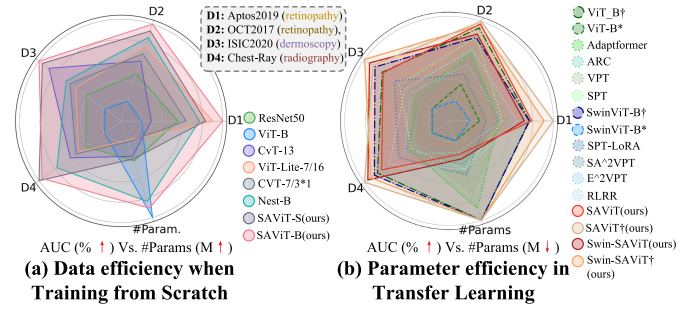


Fig. 1. Comparison with recent data and parameter efficient methods across three medical modalities. (a) SAViT demonstrates exceptional performance when trained from scratch on limited medical images, highlighting its **data efficiency** with large trainable parameters. (b) SAViT achieves excellent results with extremely lower trainable parameters, showing its **parameter efficiency** in transfer learning. †: full fine-tuning; *: linear probing.

SwinSAViT. Extensive experiments are performed. The obtained results well demonstrate the strong generalization capability of SAViT across three common medical modalities, achieving desired data and parameter efficiency. Specifically, SAViT-B improves the data efficiency of ViT-B by approximately +10% AUC scores with similar training parameters (SAViT-B: 75.87M, ViT-B: 86.60M) when trained from scratch (Fig.1(a)). In the transfer learning setting, by using only 0.17M/0.23M parameters, our parameter-efficient versions, SAViT/SwinSAViT, achieve performance comparable to fully fine-tuned ones requiring 86.60M and 88.00M parameters (Fig.1(b)).

We intend to make the following novel contributions to the field of medical image analysis:

- 1) We propose the SPA module, which innovatively integrates a visual prompt, termed Anat-Prompt, into a circular feature interaction strategy to share consistent attention across local anatomical regions in ViTs, thus enhancing the translational invariance of local features;
- 2) We introduce the MixPool module after SPA to extract hierarchical global features via multi-scale AvgPooling, thus preserving the global modeling capability of ViTs;
- 3) We design the Lr-MSA module to reduce drastically the number of parameters from the existing ViT models without performance degradation. This module serves as a lightweight, trainable adapter for transfer learning, enabling efficient parameter tuning; and
- 4) We establish a general framework named SAViT by unifying SPA, MixPool, and Lr-MSA to enhance both data and parameter efficiency in medical image analysis.

II. RELATED WORK

A. Data Efficiency in ViT

ViTs have proven effective in capturing long-range dependencies to model global relationships among images [2], [6], [7]. However, they are typically data-hungry, requiring large datasets to develop the inductive biases necessary for local feature extraction. Improving their data efficiency, i.e., enabling them to perform well on small datasets, is highly sought with the following three strategies:

1) *Knowledge Distillation*: This technique aims to transfer knowledge from larger, more complex models to smaller ones, thus enhancing learning efficiency and allowing smaller models to achieve high performance with fewer data [23], [24], [25]. For example, Lungu-Stan et al. [7] propose a skin cancer classification method by distilling the knowledge of a vanilla ViT into a smaller model, while achieving performance comparable to the full model.

2) *CNN-ViT Hybrids*: Combining CNNs and ViTs in a joint architecture is a prevalent strategy to enhance data efficiency. A notable example is CvT [3]. It enhances local feature extraction by using convolutional token embedding and linear projection before each self-attention block. Following this, CVT-CCT [26] adapts the CNN-ViT joint model for small-scale datasets, achieving success on CIFAR10 and CIFAR100 but facing limitations in medical analysis due to inadequate consideration of anatomical information. In the medical domain, H2Former [27] integrates a hierarchical hybrid of convolution, multi-scale channel attention, and transformer features for tumor segmentation. M-CnT [6] learns manifold structures across organs with CNNs and transformers. Kuang et al. [8] use a circular module for accurate stroke lesion segmentation with CNN and transformer encoders. CKD-TransBTS [28] uses a cross-attention block to extract multi-modality image features from CNNs and ViTs. Despite their effectiveness, these complex CNN and ViT hybrids may be challenging to optimize for diverse medical modalities.

3) *Hierarchical Models*: They use multi-level structure [22] or multi-scale patch embedding [21] to balance efficiency and detailed visual feature capture. While these lightweight ViTs reduce reliance on large pre-trained datasets like ImageNet-21k and JFT-300M [21], [22], they still require medium-scale datasets like ImageNet1K for optimal performance. On small-scale datasets like CIFAR, the hierarchical model NestNet [4] shows significant performance improvements over CNNs when training ViTs from scratch by using block aggregation and cross-block communication. However, exploring this approach in the medical domain remains limited.

In SAViT, we thoroughly investigate visual prompt learning that assists ViTs in capturing fine-grained features from limited medical data based on anatomical consistency. Unlike previous data-efficient models [3], [26] based on complex network architectures, the prompt in our SAViT is plug-and-play and able to self-update to acquire local information without requiring additional modality knowledge, thereby ensuring generalization across various medical analyses.

B. Parameter Efficiency in ViT

Recent advancements in parameter efficiency for ViTs have been significantly driven by methods such as Adapter and prompt learning. These approaches offer flexible and scalable solutions for adapting pre-trained models to new tasks while minimizing computational and memory overhead.

1) *Adapters*: They are small, trainable modules inserted into pre-trained models, allowing adaptation to new tasks with minimal modifications to the original parameters [29]. A notable technique, LoRA [30], adapts pre-trained models by introducing low-rank matrices into the existing layers, enhancing

parameter efficiency compared to full fine-tuning. Adaptformer [29] uses adapters within the Multi-Layer Perceptron (MLP) layer of ViTs. ARC [31] is a parameter-sharing scheme that can create layer-adaptive adapters. SPT-LoRA [32] adaptively allocates adapters to task-specific important positions given a desired tunable parameter budget. RLRR [33] fine-tunes the frozen pre-training parameter matrix via a residual-based low-rank rescaling strategy. While Adapters have demonstrated to be effective in various medical applications, such as medical visual question answering [34] and segmentation tasks [15], their potential for handling fine-grained medical anatomy remains underexplored.

2) *Prompt Learning*: In contrast to Adapters, prompts are specific embeddings inserted into the token space to guide attention mechanisms for new tasks. Recent prompt-based studies have increasingly focused on vision-language models [35], [36], with their application remaining largely confined to text encoders [37]. Since the advent of ViT, visual prompt learning has been investigated as a method for fine-tuning a model across various downstream vision tasks [11], [12], [13], [14], [38]. For example, Visual Prompt Tuning (VPT) [11] introduces learnable prompt tokens before input image tokens. Self-Prompt Tuning (SPT) [12] improves this by initializing prompts with downstream token prototypes to capture richer task information. Visual Fourier Prompt Tuning [38] incorporates Fast Fourier Transform-based embeddings to fuse spatial and frequency features for efficient adaptation. In the medical domain, research has predominantly centered on text prompts derived from models like GPT [37] or medical reports [39]. While some studies have explored the use of visual prompts for medical applications, including domain adaptation [40] and segmentation [16], [17], further research is needed to fully explore their efficiency and adaptability across different medical domains.

3) *Reparameterization*: These methods aim to reduce computational cost by tuning parameters that are inherently in, or can be reparameterized into, the backbone during inference [41], [42], [43]. Recently, representation tuning has emerged as a new approach under this paradigm. RED [41] performs direct representation editing through element-wise scaling and biasing across Transformer layer representations, while ReFT [42] applies intervention-based tuning using low-rank orthonormal projections. Although effective in single-modality (language) settings, representation tuning for multimodal models remains underexplored. MRT [43] represents an early attempt to extend intervention-based representation tuning to multimodal scenarios through structured design.

In our work, by considering anatomical structure consistency and redundancy, we design a combined adapter-prompt training scheme that balances accuracy and computational demands, thereby ensuring parameter efficiency in both transfer learning and training from scratch scenarios.

III. PROPOSED METHOD

A. Overview

We propose the Shared Attention Vision Transformer (SAViT) to enhance both data and parameter efficiency of

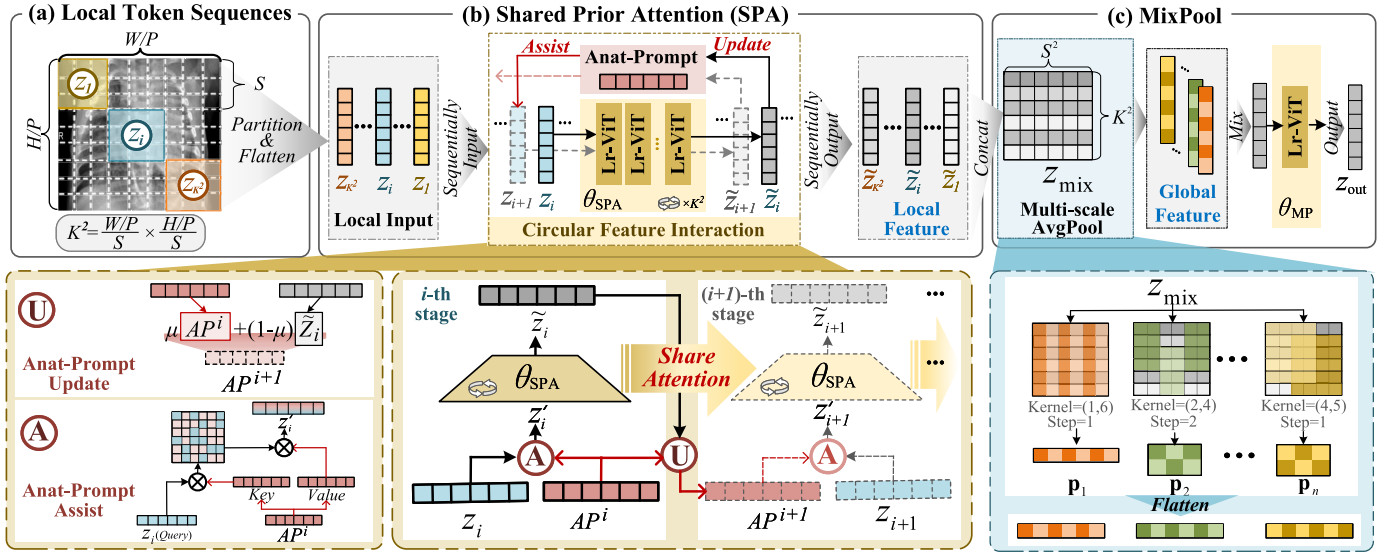


Fig. 2. The overall architecture of SAViT. (a) We first partition the entire token map into K^2 local token sequences, denoted as $\{Z_i\}_{i=1}^{K^2}$. (b) We propose the SPA module that computes each Z_i sequentially by using shared parameters (θ_{SPA}) in a Circular Feature Interaction Strategy. In this module, Anat-Prompt (AP^i) is designed as a learnable visual embedding that progressively integrates prior anatomical knowledge, assisting in the interaction with the input local features (Z_i) to enhance translation invariance. (c) We introduce MixPool that extracts hierarchical global features from the outputs of SPA, preserving long-range dependencies in SAViT. Lr-ViT incorporates Low-rank MSA that is proposed for parameter efficiency in medical image analysis. Details of Lr-ViT are provided in Fig. 3.

ViTs in medical image analysis. The SAViT architecture, shown in Fig. 2, includes: i) Shared Prior Attention (SPA) that introduces Anat-Prompt in a circular feature interaction strategy to enhance translation invariance for data efficiency; ii) MixPool that extracts hierarchical global features from SPA outputs, preserving long-range dependencies; and iii) Low-rank Multi-head Self-Attention (Lr-MSA) in ViTs (Lr-ViT) that reduces parameters and can be adopted as a trainable adapter in frozen ViTs, thus enabling parameter-efficient transfer learning. SAViT is mathematically formulated as:

$$\mathcal{Z} = \text{GeoPart}(\mathbf{I}) = \{Z_i | i = 1, \dots, K^2\}, \quad (1)$$

$$\tilde{Z}_i = \text{SPA}(\text{Lr-ViT}(Z_i, AP^i)), \quad (2)$$

$$Z_{out} = \text{MixPool}(\tilde{Z}_1, \tilde{Z}_2, \dots, \tilde{Z}_{K^2}), \quad (3)$$

where GeoPart is Geometric Partition, $\mathbf{I}$ represents the image token map following position embedding, as defined in [1]. $\mathcal{Z}$ is the set of local token sequences corresponding to various local anatomical regions. K^2 is the count of Z_i in $\mathcal{Z}$, with each Z_i sharing the same parameters in SPA. AP^i denotes Anat-Prompt, a learnable visual embedding with the same dimensionality as Z_i 's. $\tilde{Z}_i$ is the local output of SPA. Z_{out} is the final extracted feature from SAViT used for prediction. We detail SPA, MixPool, and Lr-ViT next.

B. Shared Prior Attention (SPA)

The data inefficiency of ViTs is primarily attributed to their lack of local inductive biases such as translation invariance [4], [44]. To address this issue in the medical domain, we propose SPA. It first geometrically partitions the entire image token map into non-overlapping local token sequences as shown in Fig. 2(a). Next, we design a circular feature interaction strategy

as shown in Fig. 2(b) to share identical ViT attention across local token sequences, aided by the learnable Anat-Prompt. This prompt sequentially encodes prior anatomical knowledge from different image regions, thus facilitating the interaction of region-invariant information and enhancing translation invariance.

1) Geometric Partition of Local Token Sequences: To prioritize ViTs focusing on local information and effectively provide prior knowledge from one anatomical region to another, we partition the canonical image token map into a set of local token sequences that align with the image's geometric structure. As shown in Fig. 2(a), given an input image of shape $\mathbb{R}^{H \times W \times C}$, the token map, after position embedding, is arranged into a grid of shape $\mathbf{I} \in \mathbb{R}^{\frac{H}{P} \times \frac{W}{P} \times d}$, where P is the patch size and d is the token dimension. These tokens are then grouped into K^2 non-overlapping local token regions, denoted as $\mathcal{Z} = \{Z_i\}_{i=1}^{K^2}$. Tokens within each region are flattened to form $Z_i \in \mathbb{R}^{S^2 \times d}$, where S^2 is the number of tokens in Z_i . Consequently, the size of $\mathcal{Z}$ is given by $K^2 = \frac{W/P}{S} \times \frac{H/P}{S}$, where $W = H$ generally.

2) Circular Feature Interaction Strategy: To incorporate local information in ViTs, existing hybrid methods process the window-wise token sequences in parallel [22], [45], [46]. While being effective, they often focus on feature interactions within immediate surroundings, neglecting the accumulation of spatial context crucial for understanding anatomical structures. To address this problem, we propose a Circular Feature Interaction strategy (CFI). It iteratively accumulates anatomical semantic information from each local anatomical region, assisted by Anat-Prompt. This strategy enables the integration of anatomical knowledge from previous iterations into the current local feature extraction process, allowing the model to learn region-specific details while preserving anatomical spatial relationships.

Specifically, as depicted in Fig. 2(b), Z_i is processed sequentially through SPA, sharing the same parameter set θ_{SPA} . This recursive process of Z_i and Z_{i+1} to share θ_{SPA} can be formulated as:

$$Z_{i+1} = f_{\text{SPA}}(f_{\text{A}}(Z_i, AP^i); \theta_{\text{SPA}}), \quad i = 1, \dots, K^2 - 1. \quad (4)$$

Here, Z_i denotes the i -th local token sequence, f_{SPA} represents the Lr-ViT blocks in SPA, and f_{A} describes the assistant process by which the Anat-Prompt, AP^i , interacts with Z_i . Since the same θ_{SPA} is applied uniformly across all locations of the input image, the network's response to a feature translates consistently as the region shifts across the image. This process effectively captures the translation invariance of region-specific details across different anatomical regions.

3) Anatomical-Assisted Prompt: To enhance the continuity and integrity of critical information across boundaries between local token sequences, we propose Anat-Prompt in SPA. Different from the conventional methods [16], [40], [47], [48], which designs prompts with additional annotations such as external expert knowledge or handcrafted rules, the proposed Anat-Prompt is self-updating without supervision. Specifically, by leveraging the anatomical consistency and interdependencies among various regions in medical images, we define a learnable visual embedding, termed Anat-Prompt, denoted as $AP \in \mathbb{R}^{S^2 \times d}$. It is designed to propagate anatomical knowledge from previous iterations to the current one. For the i -th iteration of CFI, we implement the exponential moving average (EMA) to update AP^i as shown in Fig. 2(b). This update process can be formulated as:

$$f_{\text{U}}(AP^i, \tilde{Z}_i) = \mu AP^i + (1 - \mu) \tilde{Z}_i, \quad (5)$$

where μ is a momentum coefficient following the cosine schedule from 0.5 to 0.9 to enrich anatomical information of AP gradually during iterations. Therefore, in the i -th iteration, AP^i can be defined as:

$$AP^i = \begin{cases} \tilde{Z}_i, & i = 1 \\ f_{\text{U}}(AP^{i-1}, \tilde{Z}_{i-1}), & i \in \{2, \dots, K^2\}, \end{cases} \quad (6)$$

For the process of prompt assistance, we employ cross-attention to integrate the prior anatomical knowledge from AP^i into Z_i . As shown in Fig. 2(b), the local token sequence Z_i directly serves as the *Query* vector, denoted as $\mathbf{Q}_{Z_i}$. The *Key* and *Value* vectors are obtained by projecting AP^i through a linear layer, where $\mathbf{K}_{AP^i} = W_{\mathbf{K}}(AP^i)$, and $\mathbf{V}_{AP^i} = W_{\mathbf{V}}(AP^i)$. Here, $W_{\mathbf{K}}, W_{\mathbf{V}}$ are shared linear maps. Therefore, cross-attention of Z_i and AP^i can be obtained as:

$$f_{\text{A}}(Z_i, AP^i) = \text{Softmax} \left(\frac{\mathbf{Q}_{Z_i} \times (\mathbf{K}_{AP^i})^{\top}}{\sqrt{d}} \right) \times \mathbf{V}_{AP^i}. \quad (7)$$

Summarily, for the i -th iteration in SPA, assisted by Anat-Prompt, input Z'_i can be computed as:

$$Z'_i = \begin{cases} Z_i, & i = 1 \\ f_{\text{A}}(Z_i, AP^i), & i \in \{2, \dots, K^2\}. \end{cases} \quad (8)$$

4) Proof of Translation Invariance in SPA: To theoretically demonstrate that SPA introduces translation invariance, we analyze its behavior under spatial translation following [49]. Let $\mathcal{T}_{\Delta}$ denote a spatial translation acting on the local token sequence Z_i :

$$\mathcal{T}_{\Delta}(Z_i) = P_{\Delta} Z_i, \quad (9)$$

where P_{Δ} is the permutation matrix induced by a spatial shift. By combining the definition of (9) and the identification of (7) the process of prompt assistance in SPA maintains permutation equivariance under translation, leading to

$$f_{\text{A}}(\mathcal{T}_{\Delta}(Z_i), AP^i) = P_{\Delta} f_{\text{A}}(Z_i, AP^i). \quad (10)$$

Because the SPA block $f_{\text{SPA}}(\cdot; \theta_{\text{SPA}})$ is parameter-shared and spatially symmetric, it preserves permutation equivariance:

$$f_{\text{SPA}}(P_{\Delta} X; \theta_{\text{SPA}}) = P_{\Delta} f_{\text{SPA}}(X; \theta_{\text{SPA}}). \quad (11)$$

Here, X denotes the input feature fed into the SPA module. Therefore,

$$\begin{aligned} f_{\text{SPA}}(f_{\text{A}}(\mathcal{T}_{\Delta}(Z_i), AP^i); \theta_{\text{SPA}}) \\ &= f_{\text{SPA}}(P_{\Delta} f_{\text{A}}(Z_i, AP^i); \theta_{\text{SPA}}) \\ &= P_{\Delta} f_{\text{SPA}}(f_{\text{A}}(Z_i, AP^i); \theta_{\text{SPA}}) \\ &= \mathcal{T}_{\Delta}(f_{\text{SPA}}(f_{\text{A}}(Z_i, AP^i); \theta_{\text{SPA}})). \end{aligned} \quad (12)$$

This equality shows that the attention in SPA response shifts consistently with the translation of the input, establishing translation invariance.

C. MixPool

To preserve long-range dependencies in SAViT, we propose a MixPool module, which builds upon SPA to extract hierarchical global information. As shown in Fig. 2(c), each local output feature $\tilde{Z}_i$ obtained by SPA is concatenated to form a feature map, formulated as $\{\tilde{Z}_i \in \mathbb{R}^{S^2 \times d}\}_{i=1}^{K^2} \xrightarrow{\text{Concat}} Z_{\text{mix}} \in \mathbb{R}^{S^2 \times K^2 \times d}$. Next, a multi-scale average pooling operation (Fig. 2(c)) is applied to Z_{mix} , generating a set of hierarchical global representations with varying levels of granularity by using receptive fields and strides of different scales. This process can be formulated as:

$$\mathbf{p}_n = \text{AvgPool}_n(Z_{\text{mix}}), \quad n \in \{1, 2, \dots, N\} \quad (13)$$

where $\mathbf{p}_n \in \mathbb{R}^{S^2 \times d}$ is the hierarchical global feature maps after the n -th AvgPool operation. N is the total number of global feature maps. Assuming that the n -th pooling kernel has dimensions (x^n, y^n) and a step size of t^n , we ensure that variables x^n , y^n , and t^n satisfy the equation $S^2 = \frac{S^2 - x^n + t^n}{t^n} \times \frac{K^2 - y^n + t^n}{t^n}$, as illustrated by the three samples in Fig. 2(c). Hence, the length of each hierarchical global feature is guaranteed to be S^2 . Then, we mix these hierarchical features as a single feature to feed it into an Lr-ViT block (θ_{MP}) within MixPool:

$$Z_{\text{out}} = f_{\theta_{\text{MP}}} \left(\sum_{n=1}^N (\text{Flatten}(\mathbf{p}_n)) \right). \quad (14)$$

This process establishes long-range dependencies across distinct regions of image, improving the model's performance in

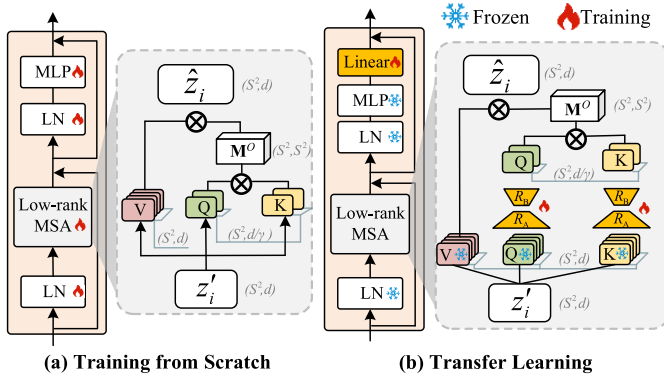


Fig. 3. Architecture of one Low-rank Vision Transformer Block. Two versions are introduced. (a) The version of training from scratch. We directly train a low-rank attention matrix in MSA. (b) The version of transfer learning. We freeze the ViT block while applying three additional linear transformations to adapt the ImageNet-pre-trained parameters.

handling complex spatial relationships. The extracted feature Z_{out} encapsulates both local and global information, enhancing data efficiency for the final prediction.

D. Low-Rank Multi-Head Self-Attention

To mitigate domain shift and enhance parameter efficiency, we propose Lr-MSA in the ViT architecture. Unlike the canonical application of Low-Rank Adaptation (LoRA) [30], which reduces computational complexity in transfer learning by introducing input/output size-invariant trainable rank decomposition matrices [32], [33], the proposed Lr-MSA using lower output dimension provides a more efficient solution in the medical domain. Underlying idea is that retaining high dimensionality in all vectors of *Query*, *Key*, and *Value* (QKV) might be excessive for capturing essential diagnostic features. Since non-diagnostic information is often redundant, Lr-MSA reduces the dimension of **Q** and **K** to focus on critical information while preserving the high dimensionality of **V** for fine-grained details. Specifically, we formulate the process of $\hat{Z}_i = \text{Lr-MSA}(Z'_i)$ as follows:

$$(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = (Z'_i \mathbf{U}^q, Z'_i \mathbf{U}^k, Z'_i \mathbf{W}^v), \quad (15)$$

$$\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{S^2 \times \frac{d}{\tau}}, \quad \mathbf{V} \in \mathbb{R}^{S^2 \times d}, \quad (16)$$

$$\mathbf{M}^O = \mathbf{Q} \times \mathbf{K}^\top, \quad (17)$$

$$\hat{Z}_i = \text{Softmax}\left(\frac{\mathbf{M}^O}{\sqrt{d}}\right) \times \mathbf{V}, \quad (18)$$

where the matrices **U** and **W** are learnable parameters, S^2 is the number of tokens in Z'_i , τ is the reduction ratio, and $\mathbf{M}^O$ is the attention matrix.

In the training from scratch scenario (Fig. 3(a)), **U** and **W** directly project $Z'_i \in \mathbb{R}^{S^2 \times d}$ into three vectors: **Q**, **K** $\in \mathbb{R}^{S^2 \times \frac{d}{\tau}}$ and **V** $\in \mathbb{R}^{S^2 \times d}$. For the transfer learning scenario (Fig. 3(b)), we adapt the LoRA approach [30] by applying rank decomposition matrices to adjust the dimensions of **Q** and **K**. Here, $\mathbf{W}^v$ represents the frozen pre-trained weight, while $\mathbf{U}^q$ and $\mathbf{U}^v$ include both frozen pre-trained weights and additional trainable components: $R_A \in \mathbb{R}^{d \times \gamma}$ and $R_B \in \mathbb{R}^{\gamma \times \frac{d}{\tau}}$

with $\gamma = 8$ empirically. To adapt the output of the frozen MLP, a trainable linear layer is introduced for better performance.

IV. EXPERIMENTS

A. Implement Details

1) **Datasets:** We conduct experiments on medical datasets spanning distinct medical imaging modalities: i) Retinopathy: Aptos2019 [50] and OCT2017 [51]; ii) Dermoscopy: ISIC2020 [52]; iii) Pathology: MHIST [53]; and iv) Radiography: Chest-XRay [51], CXP [54], SIIM-ACR,¹ and NIH [55]. Four tasks are evaluated: multi-class classification on Aptos2019 and OCT2017, binary-class classification on MHIST, ISIC2020 and Chest-XRay, multi-label classification on CXP, and weakly-supervised localization on NIH. Details of these datasets are presented in [56].

2) **Evaluation Metrics:** Model performance for classification is reported by using the Area Under the Receiver Operating Characteristic Curve (AUC), while localization performance is measured using Intersection over Union (IoU). During the analysis, we employ data augmentation techniques, including RandAugment, CutMix, and MixUp, to mitigate the label imbalance. For each dataset, we perform five-fold cross-validation to ensure robust analysis. Each experiment was repeated three times with different random seeds, and we report the average score along with the standard deviation.

3) **Model Settings:** In our notation, the number of Lr-ViT blocks in SPA is flexible. Therefore, we denote two sizes of SAViT to evaluate the data efficiency. Namely, i) SAViT-S consists of 2 Lr-ViTBlocks in SPA and 1 Lr-ViTBlock in MixPool, where patch size $P = 16$, token dimension $d = 768$, reduction ratio $\tau = 8$, the size of local token sequence $S = 7$, and the input image size is $\mathbb{R}^{224 \times 224 \times 3}$. ii) SAViT-B consists of 11 Lr-ViTBlocks in SPA and 1 Lr-ViTBlock in MixPool to stack the same architecture of ViT-B. We also plug in another two types of transformer blocks in SPA and MixPool to evaluate the parameter-efficient in transfer learning, which are named SAViT (Shared Attention Vision Transformer) using ViT-B blocks [1] and Swin-SAViT (Shift-window Shared Attention Vision Transformer) using SwinViT-B blocks [22]. SAViT-B, SAViT, and Swin-SAViT have the same hyperparameters as SAViT-S has.

4) **Training Settings:** The AdamW optimizer is utilized with a weight decay of 10^{-8} under the cosine scheduler [57] during the training process. The learning rate follows a linear ramp-up for the first 10 epochs, reaching its base value according to the linear scaling rule: $lr = 0.0001 \times \text{batchsize}/256$. For both training from scratch and transfer learning, all models are trained using distributed training for most 300 epochs with a batch size of 64 on 2 NVIDIA 3090 GPUs. We employ the early stopping when the model's performance starts to degrade 8 epochs on the validation set.

B. Ablation and Analysis

We evaluate the contribution of each component in transforming ViT-B into SAViT-B, with a focus on the proposed

¹<https://www.kaggle.com/c/siim-acr-pneumothorax-segmentation>

TABLE I
COMPONENT-WISE ANALYSIS OF SAVIT. TOP-1 AUC WHEN
TRANSFORMING ViT-B INTO SAVIT-B STEP BY STEP

	SPA	Lr-MSA	MixPool	#Param. (M ↓)	OCT2017 (AUC% ↑)	ISIC2020 (AUC% ↑)	Chest-Ray (AUC% ↑)
	CFI	Anat-Prompt					
(a) Training from scratch							
1	✓	✓	✓	86.60	89.75 ± 0.69	79.77 ± 3.89	78.24 ± 3.20
2	✓	✓	✓	86.60	86.59 ± 2.67	76.28 ± 1.87	77.69 ± 0.58
3	✓	✓	✓	86.60	93.54 ± 1.02	90.40 ± 0.49	83.77 ± 0.88
4	✓	✓	✓	86.60	97.17 ± 1.08	93.01 ± 2.62	87.00 ± 0.92
5	✓	✓	✓	75.87	94.67 ± 0.97	91.58 ± 1.23	83.64 ± 1.89
6	✓	✓	✓	75.87	97.98 ± 0.26	93.68 ± 0.88	86.44 ± 0.22
(b) Transfer learning using ImageNet-pre-trained parameter							
7	✓	✓	✓	86.60	98.67 ± 0.58	95.68 ± 0.89	89.69 ± 0.35
8	✓	✓	✓	0.001	83.78 ± 0.31	70.98 ± 0.77	67.96 ± 0.58
9	✓	✓	✓	0.001	95.54 ± 0.02	92.19 ± 0.38	83.77 ± 0.88
10	✓	✓	✓	0.001	96.33 ± 0.03	94.88 ± 0.08	85.69 ± 0.27
11	✓	✓	✓	0.17	97.55 ± 0.07	94.17 ± 0.31	88.64 ± 0.75
12	✓	✓	✓	0.17	98.85 ± 0.08	95.62 ± 0.07	89.29 ± 0.07

CFI and Anat-Prompt in the SPA module, as well as Lr-MSA and MixPool. Table I reports the Top-1 AUC scores across three medical image datasets: OCT2017, ISIC2020, and Chest-Ray, each representing a different modality.

1) *Impacted Components of Data Efficiency*: The ablation study in Table I(a) begins with a baseline ViT-B model (row 1), which is trained from scratch without any proposed components. Specifically, when adding CFI alone (row 2), the SAViT model divides the entire input image into several local token sequences and processes them sequentially. In this setting, there is no interaction or contextual exchange among different local sequences. As a result, the model can only perceive partial regions of the image at each step, losing the continuity and integrity of critical anatomical structures across the boundaries of these local partitions. This disrupted holistic modeling leads to incomplete spatial understanding and, consequently, performance degradation. However, the introduction of Anat-Prompt within CFI (row 3) mitigates this issue and improves overall performance, owing to its ability to preserve continuity across local token sequences, thereby ensuring translational invariance. The MixPool module (row 4) further boosts performance by maintaining the long-range dependencies of local features. Adding Lr-MSA without MixPool (row 5) shows a slight improvement over row 3, confirming that the lower-dimensional queries and keys in SAViT do not harm performance. Combining all modules (row 6) results in the highest AUC scores, demonstrating that SPA and MixPool enhance ViT's data efficiency when trained on limited medical data, while Lr-MSA reduces computational demands without sacrificing performance.

2) *Impacted Components of Parameter Efficiency*: Table I(b) shows the ablation results of SAViT in the transfer learning setup, where pre-trained parameters are loaded and frozen, with the proposed components added incrementally. Row 7 demonstrates the performance of ViT-B with full parameter fine-tuning. When only CFI is applied (row 8), performance drops due to restricted local receptive fields, which impair long-range dependencies of the pre-trained parameters. Anat-Prompt (row 9) addresses this by acting as a semantic bridge that propagates anatomical priors and contextual information from previous sequences to subsequent ones. This mechanism enables SAViT to maintain cross-partition continuity

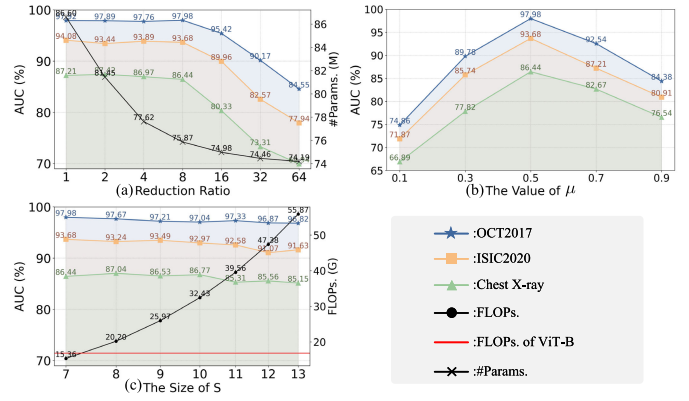


Fig. 4. (a) Analysis of the reduction ratio in Lr-MSA. We utilized the default settings in SAViT-B, except for varying the value of the reduction ratio (τ); (b) Analysis of the updating ratio in Anat-Prompt Update. We utilized the default settings in SAViT-B, except for varying the value of μ in (5); (c) Analysis of the size of local token sequences (S). We keep the value of K , except for varying the value of S in Z_i .

and to integrate global contextual cues while processing local sequences. Adding MixPool further enhances SAViT, preserving long-range dependencies. Combining Anat-Prompt with Lr-MSA Adapters (row 11) also boosts performance, confirming the efficacy of the lightweight adaptation strategy. Finally, integrating all modules (row 12) achieves performance comparable to row 1, while requiring only 0.17M trainable parameters, showcasing its parameter efficiency in transfer learning.

3) *Effectiveness of Lr-MSA*: To showcase the efficacy of Lr-MSA, we have conducted experiments to test different rates of reduction ratio (τ in (16)) for the vectors of $\mathbf{Q}$ and $\mathbf{K}$. The results are presented in Fig. 4(a). As τ increases, the parameters of SAViT consistently decrease, with only minor fluctuations in performance up to $\tau=8$. However, for values of $\tau > 8$, a sharp decline in performance is observed. This demonstrates that, for $\tau \leq 8$, the low-dimensional matrices of $\mathbf{Q}$ and $\mathbf{K}$ are capable of effectively capturing detailed and precise information for efficient medical image diagnosis. To this end, we set the reduction ratio to 8 as the default in the SAViT framework, achieving a balance between accuracy and computational efficiency.

4) *Analysis of μ in Anat-Prompt Updating*: To analyze the influence of the updating ratio μ in (5), which controls the ratio of interaction between current and previous local features, we conduct experiments by varying μ from 0.1 to 0.9. The results, presented in Fig. 4(b), show that extreme values of μ lead to performance degradation. Specifically, when $\mu=0.9$, the model overly prioritizes previous features, neglecting the importance of current features, which results in an inadequate incorporation of new anatomical information. On the other hand, when $\mu=0.1$, the model excessively focuses on current features, disregarding valuable prior anatomical knowledge. This lack of interaction among local anatomical features undermines the model's ability to capture translation invariance, which is crucial for robust performance. As a result, the model experiences performance degradation under this setting. To strike a balance, we select $\mu=0.5$, which

TABLE II

COMPARISONS OF INFERENCE TIME PER ITERATION (S/IT) AMONG ViT, SWIN-ViT AND SAViT WITH DIFFERENT SIZE OF S

ViT	SwinViT	SAViT (ours)			
		$S = 7$	$S = 9$	$S = 11$	$S = 13$
1.370 s/it	1.213 s/it	0.495 s/it	2.135 s/it	3.088 s/it	3.888 s/it

TABLE III

EFFECTIVENESS OF COMBINING ANAT-PROMPT AND LR-MSA IN TRANSFER LEARNING. '#PARAM': TRAINABLE PARAMETERS; '**': LINEAR PROBING; '†': FULL-PARAMETER TUNING

Anat-Prompt	Lr-MSA	OCT2017 AUC(↑,%)	ISIC2020 AUC(↑,%)	Chest-Ray AUC(↑,%)	#Param M(↓)
✓	ViT*	84.53 ± 0.22	85.31 ± 0.12	77.18 ± 0.21	0.001
✓	✓	96.33 ± 0.03	94.88 ± 0.08	85.69 ± 0.27	0.001
✓	✓	94.87 ± 0.09	90.27 ± 0.14	87.11 ± 0.08	0.17
✓	✓	98.85 ± 0.08	95.62 ± 0.07	89.29 ± 0.07	0.17
	ViT†	98.67 ± 0.58	95.68 ± 0.89	89.69 ± 0.35	86.6

ensures effective interaction and integration of current and previous features.

5) *Computational Complexities of SPA*: Our SPA's novelty lies in its processing non-overlapping local token sequences sequentially, without introducing additional computational complexity. Specifically, in the original ViT, the MSA block operates on the entire set of input tokens simultaneously. Given the token map $\mathbf{I} \in \mathbb{R}^{D^2 \times d}$, where D^2 is the total number of image tokens, the computational complexities of $\mathbf{I}$ in MSA is:

$$\Omega(\text{MSA}) = 4D^2d^2 + 2D^4d, \quad (19)$$

In SPA, we partition $\mathbf{I}$ to K^2 local token sequences $\{Z_i\}_{i=1}^{K^2}$, where $Z_i \in \mathbb{R}^{S^2 \times d}$ and S^2 is the number of image tokens in Z_i . Consequently, the computational complexity of a single MSA block to process K^2 local sequences Z_i in SPA is:

$$\Omega(\text{S-MSA}) = K^2(4S^2d^2 + 2S^4d). \quad (20)$$

When the local token sequences $\{Z_i\}_{i=1}^{K^2}$ are non-overlapping, we have $D = K \times S$. Since $K \geq 1$, it follows that

$$\Omega(\text{S-MSA}) \leq \Omega(\text{MSA}). \quad (21)$$

When overlap is present (i.e., $S > \frac{D}{K}$), the computational complexity of SPA exceeds that of the original ViT:

$$\Omega(\text{S-MSA}) > \Omega(\text{MSA}). \quad (22)$$

However, as shown by the experimental results in Fig. 4(c), the overlapping setting does not yield a significant performance improvement throughout the SAViT computation process. We further measure and compare the inference time of SPA with the baseline architectures (ViT and Swin-ViT) to analyze the computational complexity. The results are presented in Table II, which shows that SPA with $S = 7$ achieves inference efficiency. Therefore, we adopt a non-overlapping strategy in SPA, which enables efficient and effective processing of local token sequences without incurring unnecessary overhead.

TABLE IV

SENSITIVITY ANALYSIS OF DIFFERENT POOLING CONFIGURATIONS. MIXPOOL WITH AVERAGE POOLING ACHIEVES THE BEST OVERALL PERFORMANCE

	Max	Average	Oct-17 AUC(↑,%)	ISIC2020 AUC(↑,%)	Chest-Ray AUC(↑,%)
No Pooling	✗	✗	95.71±0.17	91.46±0.02	87.31±0.08
Single Pooling	✓	✗	96.34±0.32	93.33±0.23	87.78±0.12
	✗	✓	96.84±0.20	93.79±0.14	88.28±0.06
MixPool	✓	✗	97.94±0.04	95.31±0.11	88.99±0.10
	✗	✓	98.85 ± 0.08	95.62 ± 0.07	89.29 ± 0.07

6) *Effectiveness of Combining Anat-Prompt and Lr-MSA*: We conduct an ablation study using only Lr-MSA or Anat-Prompt in SAViT to better illustrate the effectiveness of combining both methods. The results are presented in Table III. When only Anat-Prompt is used, SAViT outperforms ImageNet-pretrained linear probing by approximately 10% in AUC across three medical classification datasets, demonstrating that Anat-Prompt effectively enhances domain adaptation and data efficiency. However, its performance remains slightly below that of full fine-tuning, as Anat-Prompt alone cannot fully compensate for the parameter gap. When only Lr-MSA is used, performance drops more noticeably compared to full fine-tuning, indicating that the model lacks explicit modeling of anatomical priors that Anat-Prompt provides. When both Prompt Learning and Lr-MSA are used together, SAViT achieves the best overall performance, confirming that the joint design effectively balances data and parameter efficiency for medical image analysis.

7) *Ablation of Pooling Settings in MixPool*: To further validate the effectiveness of MixPool and its ability to integrate local features into global context, we conduct a sensitivity analysis regarding different pooling configurations. Specifically, we compare three pooling configurations: 1) No Pooling, where all local token sequences output by SPA are directly added together without any aggregation operation; 2) Single Pooling, where the concatenated local token sequences are aggregated by either a single max or average pooling layer to produce a global representation; and 3) MixPool, which integrates multi-scale pooling to capture contextual information from different receptive fields. It is worth noting that these pooling configurations do not affect the computational complexity. The results presented in Table IV clearly show that MixPool, when using average pooling as the default pooling choice, achieves better performance than its two alternatives. These results demonstrate its effectiveness in facilitating the integration of local features into a coherent global representation and verify its efficiency and robustness in enhancing the model's ability to capture long-range dependencies.

8) *Insight Analysis*: To verify that SAViT effectively learns translational invariance of locality, we perform the insight analysis using Central Kernel Alignment (CKA) [58] to visualize the differences in spatial representational structures between SAViT and standard ViT-based models. CKA is a powerful method for measuring the similarity between feature matrices, where higher CKA values indicate greater similarity

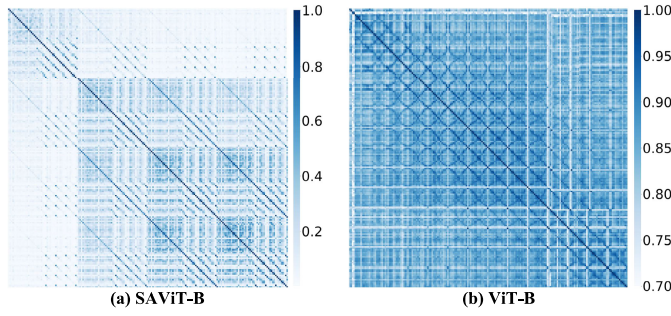


Fig. 5. Visualization of token-wise similarity. The CKA scores for all pairs of different tokens in the last layer of (a) SPA in SAViT-B and (b) ViT-B. The 'x' and 'y' axes represent the indices of image tokens.

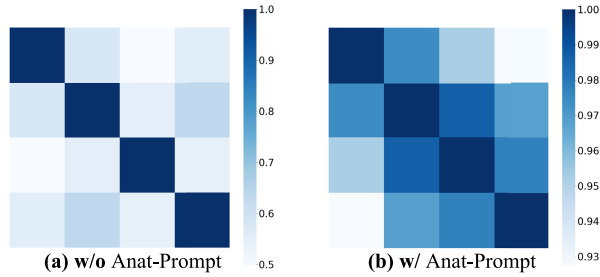


Fig. 6. Representation of Different Local Token Sequences. The CKA score was calculated for all pairs of local representations ($\tilde{Z}_i$). The 'x' and 'y' axes serve as indices for $\tilde{Z}_i$.

between the feature representations. All experiments in this section use internal representations obtained from 1000 Chest-Ray images in the Chest-Ray dataset.

a) *SAViT effectively learns translation invariance from different local anatomical regions:* We provide the insight analysis of SAViT-B and ViT-B regarding token relationships by using CKA. The results are visualized as a heatmap in Fig. 5. Compared to ViT-B, SAViT-B displays significantly lower similarity scores and a block-like structure, indicating that tokens in different positions encode distinct information, while tokens in close proximity share similar features. These findings suggest that SAViT-B captures translational invariance from local anatomical variations, which is crucial for enhancing the locality of ViTs and ensuring robust performance in medical image analysis.

b) *Anat-Prompt effectively facilitates the sharing of region-invariant information across anatomical regions:* To validate the propagation of prior anatomical knowledge across diverse local token sequences ($\tilde{Z}_i$), we quantify the CKA scores between all pairs of local representations, both with and without Anat-Prompt. Considering that the final layer comprises the MixPool module, responsible for integrating local information, we focus on and visualize the CKA scores of local representations obtained after the SPA module. Specifically, we compute the CKA scores for all pairs of local representations after SPA extraction. The results are illustrated as heatmaps in Fig. 6. When Anat-Prompt is removed (Fig. 6(a)), the similarity among diverse local representations is relatively low, reflecting significant differences across non-overlapping representations. However, these differences are effectively diminished when the proposed Anat-Prompt is

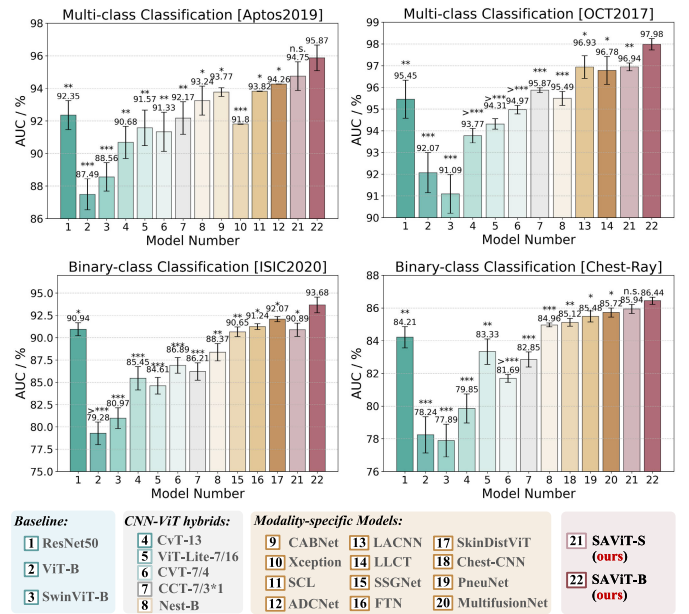


Fig. 7. Compare the proposed SAViT with other data-efficient models, where models with high AUC indicate better performance. The error bar lengths represent data uncertainty, and the center positions indicate average AUC scores. The p-values, calculated using a two-sided t-test between our SAViT-B and the other models, are presented in (a). 'n.s.': No Significance; '*': $p < 0.05$; '**': $p < 0.01$; '***': $p < 0.001$; '>****': $p < 0.0001$; #Param: trainable parameters; †: full-parameter tuning.

applied, as clearly shown in Fig. 6(b), where a smooth linear color transition from dark to light shades is observed across $\tilde{Z}_i$.

C. Comparison With State-of-the-Arts

1) *Comparison With SOTA Data-Efficient Models:* To assess the data efficiency of SAViT, we perform a comparative analysis against 20 methods trained from scratch. These methods include: i) Baselines: ResNet50 [65], ViT-B [1] and SwinViT-B [22]; ii) CNN-ViT hybrids for improving inductive biases: CvT [3], ViT-Lite [26], CVT [26], CCT [26], and Nest-B [4]; iii) Medical modality-specific methods: CABNet [5], Xception [66], SCL [67], and ADCNet [68] for Aptos2019; LACNN [69] and LLCT [70] for OCT 2017; SSGNet [71], FTN [9], and SkinDistViT [7] for ISIC2020; Chest-CNN [72], PneuNet [73], and MultifusionNet [74] for Chest-Ray. For these models, we reimplement their public code on each dataset, running experiments three times with different random seeds.

Fig. 7 presents a comparative analysis through statistical histograms. SAViT-B achieves the highest AUC scores on Aptos2019, OCT2017, ISIC2020, and Chest-Ray, significantly outperforming ViT-B and SwinViT-B ($p < 0.001$), which have similar parameter counts (75.87M for SAViT-B, 86.60M for ViT-B, and 88.00M for SwinViT-B). These results demonstrate that large ViT models can excel in the medical domain when trained from scratch with SAViT, effectively addressing ViTs' data-hungry issue. Moreover, SAViT-B surpasses SOTA modality-specific methods with significant improvements ($p < 0.05$) across all datasets, underscoring its strong generalization ability across diverse medical modalities and its data efficiency when trained from scratch. A detailed confusion matrix

TABLE V

COMPARE THE PROPOSED SWIN-SAViT/SAViT WITH OTHER PARAMETER-EFFICIENT MODELS, WHERE MODELS THAT ACHIEVE BOTH HIGH AUC AND LOW PARAMETER COUNT DEMONSTRATE EFFICIENT PERFORMANCE. THE BEST PERFORMANCE IS BOLD AND THE SECOND BEST IS UNDERLINED. ‘#PARAM’: TRAINABLE PARAMETERS; ‘*’: LINEAR PROBING; ‘†’: FULL-PARAMETER TUNING

Methods	Aptos2019 AUC(†,%)	OCT2017 AUC(†,%)	ISIC2020 AUC(†,%)	Chest-Ray AUC(†,%)	#Param. M(↓)
Parameter-efficient Fine-tuning					
ViT-B*	76.67 ± 0.26	84.53 ± 0.22	85.31 ± 0.12	77.18 ± 0.21	0.001
SwinViT-B*	78.76 ± 0.11	86.26 ± 0.11	82.38 ± 0.11	76.15 ± 0.10	0.001
Adaptformer	89.16 ± 0.12	93.95 ± 0.15	91.44 ± 0.09	82.44 ± 0.08	1.25
ARC	94.17 ± 0.08	89.23 ± 0.09	90.03 ± 0.08	83.76 ± 0.09	0.25
VPT	87.71 ± 0.09	92.79 ± 0.18	93.14 ± 0.09	85.17 ± 0.09	0.85
SPT	90.57 ± 0.14	94.90 ± 0.11	92.49 ± 0.08	86.27 ± 0.10	0.36
SPT-LoRA	91.20 ± 0.10	95.23 ± 0.05	90.88 ± 0.10	86.53 ± 0.08	0.44
SA ² VPT	88.23 ± 0.08	90.66 ± 0.07	93.21 ± 0.11	84.73 ± 0.12	0.26
E ² VPT	92.42 ± 0.08	93.93 ± 0.06	91.74 ± 0.07	83.02 ± 0.11	<u>0.18</u>
RLRR	93.74 ± 0.08	91.06 ± 0.09	90.19 ± 0.09	88.49 ± 0.06	0.33
EPT	93.65 ± 0.02	98.16 ± 0.04	91.15 ± 0.01	86.21 ± 0.03	0.54
CUFIT	90.81 ± 0.32	93.57 ± 0.56	92.22 ± 0.24	87.79 ± 0.67	2.50
SAViT (ours)	<u>96.02 ± 0.32</u>	<u>98.85 ± 0.08</u>	<u>95.62 ± 0.07</u>	<u>89.29 ± 0.07</u>	0.17
Swin-SAViT (ours)	<u>96.32 ± 0.08</u>	<u>98.23 ± 0.08</u>	<u>96.51 ± 0.05</u>	<u>90.66 ± 0.09</u>	0.23
Full-parameter Fine-tuning					
ViT-B†	95.84 ± 0.26	98.74 ± 0.15	95.71 ± 0.15	89.62 ± 0.16	86.60
SwinViT-B†	95.79 ± 0.11	98.01 ± 0.08	96.21 ± 0.09	90.07 ± 0.14	88.00
SAViT† (ours)	97.69 ± 0.12	99.15 ± 0.07	96.43 ± 0.07	90.96 ± 0.15	86.60
Swin-SAViT† (ours)	<u>97.17 ± 0.35</u>	<u>98.85 ± 0.07</u>	<u>96.83 ± 0.07</u>	<u>90.36 ± 0.12</u>	88.00

analysis, which is provided in [56], further highlights SAViT’s robust performance.

2) *Comparison With SOTA Parameter-Efficient Models:* We conduct transfer learning experiments using two training schemes: i) full-parameter tuning and ii) efficient tuning with joint Anat-Prompt and Lr-MSA. In scheme i, pre-trained ImageNet parameters are integrated into SAViT and Swin-SAViT, making all parameters trainable (denoted as SAViT† and Swin-SAViT†). In scheme ii, we freeze the pre-trained parameters and train only the R_A and R_B in Lr-MSA, updating Anat-Prompt using EMA.

a) *SAViT/Swin-SAViT outperforms other parameter-efficient models in medical image analysis:* In this comparative analysis, we evaluate both baselines and SOTA efficient-tuning models, including i) adapter-based models: Adaptformer [29], ARC [31], SPT-LoRA [32], RLRR [33], and CUFIT [62]; and ii) prompt-based models: VPT [11], SPT [12], E²VPT [14], SA²VP [13], and EPT [61]. Here, Adaptformer, ARC, VPT, SPT, CUFIT, and EPT utilize the ViT backbone, whereas SPT-LoRA, RLRR, E²VPT, and SA²VP utilize the Swin-ViT backbone. Specifically, CUFIT and EPT are designed for medical image analysis. As illustrated in Table V, the proposed SAViT/Swin-SAViT requires only 0.17M/0.23M parameters for tuning, while achieving performance comparable to fully tuned ViT-B/SwinViT-B models (86.60M/88.00M). Furthermore, compared to SOTA medical parameter-efficient approaches, such as CUFIT and EPT, SAViT/Swin-SAViT also demonstrates a significant performance improvement ($P < 0.05$), highlighting its ability to adaptively accommodate diverse ViT architectures, reduce computational demands, and achieve impressive performance in medical image analysis.

b) *SAViT/Swin-SAViT achieves more precise results than medical-pretrained models:* We have conducted multi-label classification on the CXP dataset, comparing SAViT/Swin-SAViT with four additional medical-pretrained models: Gloria [59],

TABLE VI

COMPARISONS OF MULTI-LABEL CLASSIFICATION. THE AVERAGE AUC SCORE ↑ (%) ON THE CXP DATASET ARE REPORTED. THE BEST PERFORMANCE IS BOLD AND THE SECOND BEST IS UNDERLINED. † DENOTES THE FULL-PARAMETER TUNING

Methods	Atel.	Card.	Edem.	Cons.	PE.	Mean
ViT-B† [11]	80.60 ± 0.93	83.27 ± 0.71	90.56 ± 0.43	<u>91.72 ± 0.56</u>	90.36 ± 0.48	87.30 ± 0.72
SwinViT-B† [22]	82.18 ± 1.00	85.27 ± 0.15	89.92 ± 0.76	<u>87.13 ± 1.06</u>	87.98 ± 0.92	87.50 ± 0.12
Gloria [59]	81.77 ± 0.55	84.36 ± 0.46	92.21 ± 0.41	90.29 ± 0.78	91.17 ± 0.14	87.96 ± 0.73
Popar [60]	82.18 ± 0.97	86.54 ± 0.28	90.31 ± 0.62	88.98 ± 0.52	94.33 ± 0.58	88.47 ± 0.88
DiRA [18]	82.26 ± 0.65	81.63 ± 0.47	92.16 ± 0.23	91.59 ± 0.94	93.33 ± 1.03	88.19 ± 0.54
Adam [19]	80.54 ± 0.61	83.27 ± 0.32	90.32 ± 0.83	92.63 ± 0.94	90.75 ± 0.24	87.50 ± 0.65
Adaptformer [29]	80.39 ± 0.73	78.97 ± 0.96	86.94 ± 0.15	91.26 ± 0.57	85.94 ± 0.89	84.70 ± 0.59
ARC [31]	79.92 ± 0.77	80.89 ± 0.65	88.03 ± 0.69	88.92 ± 0.42	88.27 ± 0.23	85.21 ± 0.86
SPT-LoRA [32]	79.64 ± 1.02	83.94 ± 0.98	91.27 ± 0.45	86.03 ± 0.93	89.99 ± 0.71	86.17 ± 0.22
RLRR [33]	80.37 ± 0.74	82.99 ± 0.47	90.15 ± 1.09	85.27 ± 0.31	90.16 ± 0.51	85.86 ± 0.98
VPT [11]	78.23 ± 0.79	82.21 ± 0.32	89.62 ± 0.82	87.29 ± 0.45	88.99 ± 0.15	85.27 ± 0.34
SPT [12]	77.59 ± 0.83	81.88 ± 0.44	88.92 ± 0.63	89.07 ± 0.72	87.21 ± 0.27	84.93 ± 0.74
E ² VPT [14]	78.91 ± 0.88	83.12 ± 0.79	87.49 ± 0.54	86.14 ± 0.89	90.33 ± 0.73	85.20 ± 0.77
SA ² VP [13]	79.77 ± 0.71	83.31 ± 0.26	89.94 ± 0.96	86.98 ± 0.67	89.59 ± 0.19	85.92 ± 0.42
EPT [61]	82.02 ± 0.29	84.01 ± 0.22	90.84 ± 0.99	87.68 ± 0.63	90.30 ± 0.21	86.97 ± 0.29
CUFIT [62]	81.75 ± 0.79	84.50 ± 0.64	86.10 ± 0.88	88.20 ± 0.87	86.55 ± 0.39	85.42 ± 0.98
SAViT† (ours)	83.64 ± 0.50	85.49 ± 0.24	93.55 ± 0.68	89.98 ± 0.87	93.46 ± 0.94	89.22 ± 0.83
SAViT (ours)	82.49 ± 0.14	84.98 ± 0.47	94.76 ± 0.57	90.91 ± 0.62	91.23 ± 0.51	88.87 ± 0.15
Swin-SAViT† (ours)	<u>83.97 ± 0.31</u>	<u>86.04 ± 0.29</u>	<u>93.87 ± 0.63</u>	88.09 ± 0.12	<u>94.29 ± 0.65</u>	<u>89.05 ± 0.39</u>
Swin-SAViT (ours)	82.29 ± 0.81	84.33 ± 0.30	92.89 ± 0.44	87.24 ± 0.92	92.94 ± 0.58	87.94 ± 0.28

TABLE VII

COMPARISONS WITH GENERAL MEDICAL FOUNDATION MODELS. THE AVERAGE AUC SCORE (%) IS PRESENTED. COMPARED TO MEDICAL BASELINE, ‘† *’ INDICATES A STATISTICALLY SIGNIFICANT IMPROVEMENT ($P < 0.05$) AND ‘n.s.’ DENOTES NO SIGNIFICANT DIFFERENCE. † DENOTES THE FULL-PARAMETER TUNING

Methods	Aptos2019	OCT2017	ISIC2020	Chest-Ray	CXP
RadImageNet† [63]	96.66 ± 0.11	97.88 ± 0.04	96.64 ± 0.21	91.07 ± 0.17	87.96 ± 0.16
+SAViT	96.71 ± 0.13 ^{n.s.}	97.27 ± 0.11 ^{n.s.}	96.04 ± 0.33 ^{n.s.}	90.89 ± 0.22 ^{n.s.}	88.33 ± 0.10 [*]
+SAViT†	97.31 ± 0.14[*]	99.19 ± 0.25[*]	97.97 ± 0.06[*]	91.98 ± 0.21[*]	88.99 ± 0.31[*]
LVM-Med† [64]	97.09 ± 0.06	98.96 ± 0.03	97.09 ± 0.31	90.54 ± 0.06	87.65 ± 0.22
+SAViT	97.36 ± 0.22 ^{n.s.}	98.09 ± 0.15 ^{n.s.}	97.21 ± 0.17 ^{n.s.}	90.17 ± 0.18 ^{n.s.}	88.52 ± 0.36 ^{n.s.}
+SAViT†	98.77 ± 0.13[*]	99.32 ± 0.12[*]	97.96 ± 0.14[*]	92.31 ± 0.36[*]	89.04 ± 0.11[*]

Popar [60], DiRA [18], and Adam [19]. The results in Table VI show that SAViT†/Swin-SAViT† and SAViT/Swin-SAViT significantly outperform ViT-B†/SwinViT-B†, and also surpass SOTA medical-pretrained and efficient-tuning models. Moreover, initializing SAViT with pre-trained weights from SOTA medical foundation models (RadImageNet [63] and LVM-Med [64]) results in a notable performance boost, as shown in Table VII. These improvements are attributed to the SPA and MixPool modules, which enhance local information while preserving long-range dependencies, ultimately improving performance in challenging medical tasks.

c) *SAViT improves ViT performance when integrated with medical foundation models:* To further validate the effectiveness and generalizability of SAViT beyond ImageNet-based pretraining, we extend our evaluation to recent SOTA medical foundation models. Specifically, we evaluate the performance of SAViT across three distinct medical imaging modalities by fine-tuning four modality-specific foundation models: 1) The UNI [76] and Virchow [77] models on the MHIIST dataset for pathology image classification; 2) the RetFound [78] model on the APTOS2019 dataset for fundus image classification, and 3) the PanDerm [79] model on the ISIC2020 dataset for skin lesion classification. All models employ ViT-Large backbones and are fine-tuned under parameter-efficient settings. For all experiments, we compute confusion matrices and report quantitative metrics, including AUC and Sensitivity. The results, illustrated in Fig. 8, demonstrate that integrating SAViT consistently improves model sensitivity

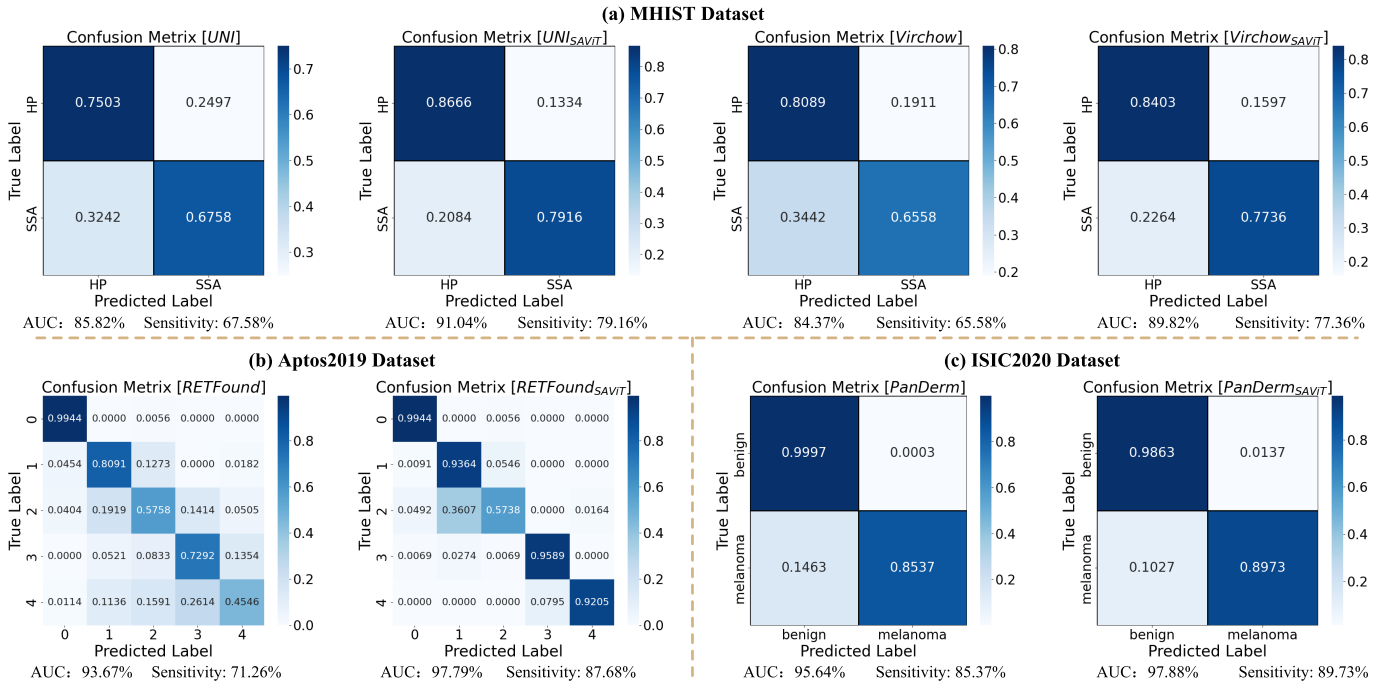


Fig. 8. Comparisons with modality-specific medical foundation models for (a) pathology, (b) retinal, and (c) dermoscopic image classification.

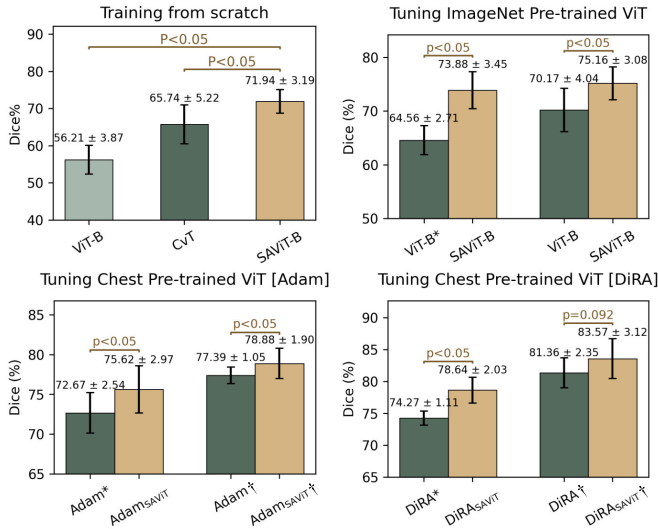


Fig. 9. Comparisons on medical image segmentation task. The error bar length represent data uncertainty, and the center positions indicate average AUC scores. The p-values, computed using a two-sided t-test between our SAVIT model and the other methods. * means decoder-only tuning, and † denotes full fine-tuning.

while maintaining comparable AUC performance, confirming its strong generalization capability and efficiency when applied to powerful medical foundation models.

d) *SAViT achieves both data and parameter efficiency of ViT on medical segmentation tasks:* To evaluate the effectiveness of our SAVIT model on medical image segmentation tasks. We conduct experiments on the SIIM-ACR dataset, which comprises more than 120,000 chest X-ray images with manually annotated pneumothorax masks. The dataset is randomly divided into training, validation, and test subsets with a ratio of

TABLE VIII

COMPARISONS OF WEAKLY-SUPERVISED LOCALIZATION ON NIH. THE AVERAGE IOU ↑ (%) AT VARYING THRESHOLDS (δ) ON THE NIH DATASET IS PRESENTED. THE BEST PERFORMANCE IS BOLD AND THE SECOND BEST IS UNDERLINED. † DENOTES THE FULL-PARAMETER TUNING

	$\delta = 0.1$	$\delta = 0.2$	$\delta = 0.3$	$\delta = 0.4$	$\delta = 0.5$	$\delta = 0.6$
ViT-B† [1]	53.17	34.87	25.43	17.88	9.21	1.36
ConvNeXt-B† [75]	54.81	36.92	26.74	17.14	10.35	2.34
DiRA† [18]	<u>58.32</u>	<u>40.85</u>	<u>28.49</u>	<u>18.92</u>	<u>12.20</u>	<u>5.42</u>
SAViT†(ours)	58.98	42.77	30.46	19.78	13.94	6.17

7:1.5:1.5. We compare different training strategies, including training from scratch, tuning ImageNet pre-trained ViT, and tuning medical-pretrained ViTs (i.e., Adam [19] and DiRA [18], both pretrained on large-scale chest X-ray datasets). We use SGD with a learning rate of 2×10^{-5} and a layer-wise decay of 0.65 across 12 transformer layers, and distribute all experiments across four NVIDIA A6000 GPUs. The results in Fig. 9 show that the proposed SAVIT consistently achieves higher Dice scores across all settings, indicating superior segmentation performance as well as improved data and parameter efficiency. Specifically, under the training-from-scratch setting, SAVIT significantly outperforms the CNN-ViT hybrid model (CvT [3]), demonstrating its superior data efficiency. Under the parameter-efficient fine-tuning mode, SAVIT not only substantially improves the performance of the ImageNet pre-trained ViT, but also achieves remarkable gains when equipped with medical-pretrained foundation models such as Adam [19] and DiRA [18]. Notably, under the full fine-tuning mode, SAVIT also achieves competitive performance compared with medical foundation models, further validating its effectiveness on medical image segmentation tasks.

e) *SAViT† excels in detecting fine-grained local disease:* We examine our SAVIT† in a weakly supervised setting, focusing

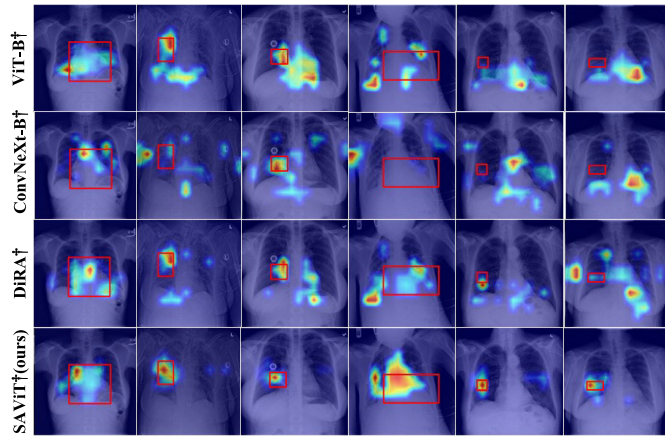


Fig. 10. Visualization of Grad-CAM disease localization in retinopathy images. The ground truth bounding box are highlighted in red.

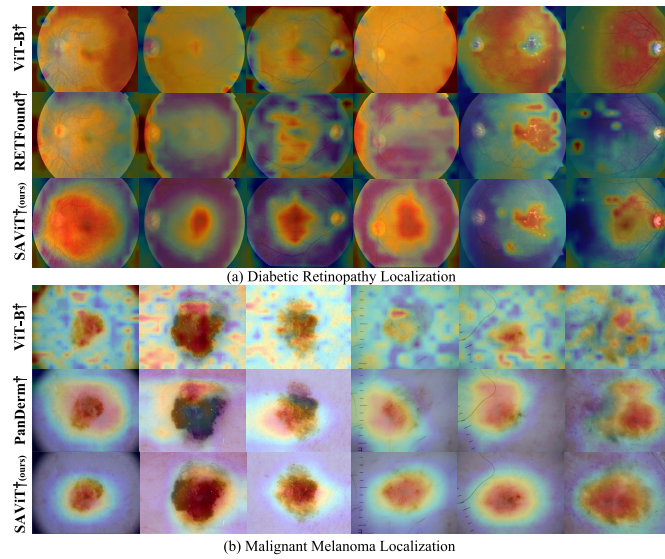


Fig. 11. Visualization of Grad-CAM on (a) the APTOS2019 dataset and (b) the ISIC2020 dataset, where SAViT better highlights the macular and posterior pole regions in fundus images, and the central pigmented and border regions of melanoma lesions in dermoscopic images.

on its efficacy in localizing thoracic pathologies. For the training process, following DiRA [18], we train SAViT† with only image-level disease labels. For the quantitative analysis, we evaluate localization accuracy using IoU between the generated bounding boxes and the ground truth, with thresholds $\geq \delta$, where δ ranges from 0.1 to 0.6. The results (Table VIII) show that SAViT† consistently outperforms its counterparts across all six IoU thresholds, highlighting the robustness of SAViT† in extracting fine-grained local information, which is crucial for accurate disease diagnosis. A detailed setting is provided in [56]. For qualitative analysis, we utilize Grad-CAM [80] to produce heatmaps that delineate the spatial distribution of specific thoracic diseases. Visualization results. The disease localization in retinopathy images, as shown in Fig. 10, indicates that the regions highlighted by SAViT† provide more accurate disease localization compared to other fine-tuned models. Furthermore, disease localization visualizations in retinopathy and dermoscopy images are presented in Fig. 11. In fundus images, SAViT†’s visualizations focus

more accurately on the macular and posterior pole regions, which are associated with typical diabetic retinopathy lesions. Similarly, in dermoscopic images, SAViT† better emphasizes the central pigmented and border regions of skin lesions, which correspond to diagnostically significant features such as irregular pigmentation and lesion boundaries, than its peers do. These Grad-CAM results demonstrate the superior effectiveness and robustness of SAViT† in detecting fine-grained local disease across various medical imaging modalities.

V. CONCLUSION

We propose a general framework called Shared Attention Vision Transformer (SAViT) to improve data and parameter efficiency in medical image analysis. Extensive experiments validate SAViT’s effectiveness in improving local information extraction, especially translation invariance, within global anatomical relationships. These benefits arise from three proposed modules: SPA adapts the Anat-Prompt to incorporate translation invariance, MixPool preserves hierarchical global information, and Lr-MSA reduces computational demands while maintaining performance. The combination of Lr-MSA and Anat-Prompt also effectively mitigates domain shift in transfer learning. Future work aims to enhance SAViT for improved compatibility with ensemble multiple-instance learning and self-supervised pre-training, as well as exploring the potential of SPA in improving the understanding of specific anatomical structures’ details. We also plan to investigate new ways [10], [81] to improve SAViT’s adaptability to unseen domains, with the goal of broadening its applicability beyond the datasets used in our current experiments, resulting in substantial real-world impact.

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